Final Evaluation Report

X Sentiment + BTC Price Forecasting | Reproducible price evaluation and honest data status | 2026-11-01

Status: software integration is implemented and tested. The real X sentiment experiment remains blocked because no eligible real posts were accepted or scored.

Scope and data

The measured dataset contains 180 Binance Vision Spot BTCUSDT daily bars covering 2026-05-05 through 2026-10-31 UTC. The quote currency is USDT, not USD. Thirty-six official archive files matched their published SHA-256 sidecars; the daily snapshot has no missing dates or duplicates. The source data and derived evaluation artifacts are licensed CC BY-NC-SA 4.0, attributed to Binance Vision, and restricted to non-commercial use.

No real X posts were included in the measured evaluation. Synthetic records are used only in deterministic software tests and are not historical data or model evidence.

Forecast evaluation

Model	Mean MAE	Relative to naive
Naive persistence	1,824.23 USDT	Baseline
Price-only ridge	2,378.50 USDT	30.38% worse
Sentiment + price candidate	2,378.50 USDT	Identical because no posts were scored

The evaluation used five expanding walk-forward folds, a five-row purge/embargo, six close horizons (n+0 through n+5), and 100 out-of-sample predictions per model and horizon, for 1,800 saved prediction rows. The candidate model result is not evidence of sentiment forecasting.

The price-only long/flat strategy reports Sharpe 1.124, Sortino 2.431, and maximum drawdown 5.885%. These are sample-specific research statistics, not a profitability claim.

X-post integration and temporal safeguards

Normalized XPostRecord batches can be merged across query modes using deterministic post-ID deduplication. Conflicting immutable identity fields fail closed. The merge sidecar preserves query and source hashes without copying post text. Event timestamps and first-retrieved timestamps are retained; point-in-time feature construction excludes future and late-retrieved posts. The optional --posts-provenance input binds the sidecar to the exact posts snapshot. Existing --posts input remains supported.

Synthetic offline unit and integration tests exercise normalization, multi-query merge, provenance, duplicate conflicts, timestamps, sentiment scoring, and leakage cutoffs. News context remains separate and does not enter predictive sentiment features. These tests validate software behavior only.

Provider verification

Provider	Verified state
Tavily	One bounded live finance search returned five results without usable publication timestamps; dates were withheld.
Exa	One bounded live search returned one result with publication time and source metadata; ongoing use and credit limit remain for employer clarification.
Apify	Authentication and Actor metadata were verified. No Actor scraping run was performed; no scraping credits were used.

Limitations and required decisions

Real sentiment forecasting has NOT been validated: zero accepted real X posts and zero scored posts. Apify historical search does not guarantee complete historical coverage. A rights-cleared, point-in-time X dataset or an employer-approved source, terms, and explicit spend cap are needed. BTCUSDT is not BTC-USD; employer acceptance or an appropriately licensed exact-USD source is needed. Clarification is also needed for the n+0 convention, the calendar deadline, and whether ongoing Exa use is required.

Coinbase Market Data was not used because its terms restrict machine-learning use without express written consent. No claims of sentiment lift, real-X model performance, or profitability are made.

Reproducibility and validation

Source-level verification on the release tree: 228 offline tests passed; Ruff, documentation/link validation, dataset registry validation, and compile checks passed. The project documents the commands in submission/REPRODUCIBILITY.md. GitHub Actions passed on the exact PR head after push.

For complete methods, row-level predictions, source attribution, data limitations, and current release references, see the repository files submission/FINAL_EVALUATION_REPORT.md, reports/btcusdt-180d/, and PR #14. This PDF does not replace the machine-readable artifacts.